

Comparing CSDID Estimators: Algorithms, Base Periods, and Software

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```
# Core DiD packages
library(did)          # Callaway & Sant'Anna (C&S) - att_gt()
library(fixest)       # Sun & Abraham via sunab()
library(staggered)    # Roth & Sant'Anna staggered estimator

# Output / helpers
library(dplyr)
library(tidyr)
library(ggplot2)
library(kableExtra)
library(patchwork)

# Stata integration (requires Stata on PATH)
# library(Statamarkdown) # uncomment if Stata is available
```

1 Data

We use the `mpdta` dataset from the `did` package — a balanced panel of county-level teen employment rates across 500 counties, where counties are treated at different calendar years (staggered adoption).

```
data(mpdta, package = "did")
```

```
# Quick overview
```

```
head(mpdta)
```

```
##      year countyreal      lpop      lemp first.treat treat
## 866 2003      8001 5.896761 8.461469      2007      1
## 841 2004      8001 5.896761 8.336870      2007      1
## 842 2005      8001 5.896761 8.340217      2007      1
## 819 2006      8001 5.896761 8.378161      2007      1
## 827 2007      8001 5.896761 8.487352      2007      1
## 937 2003      8019 2.232377 4.997212      2007      1
```

```
cat(sprintf(
  "Counties: %d | Years: %s | Treatment cohorts: %s\n",
  n_distinct(mpdta$countyreal),
  paste(sort(unique(mpdta$year)), collapse = ", "),
  paste(sort(unique(mpdta$first.treat[mpdta$first.treat > 0])), collapse = ", ")
))
```

```
## Counties: 500 | Years: 2003, 2004, 2005, 2006, 2007 | Treatment cohorts: 2004, 2006, 2007
```

The outcome is `lemp` (log teen employment), the unit ID is `countyreal`, the time variable is `year`, and `first.treat` gives the first treatment period (0 = never treated).

2 Part A: Comparing Estimation Algorithms (`did::att_gt`)

`att_gt()` from the `did` package supports three estimation methods:

Method	Description
"dr"	Doubly-Robust IPW (default) — combines outcome regression + propensity score
"ipw"	Inverse Probability Weighting only
"reg"	Outcome Regression only

```
run_attgt <- function(method, control = "nevertreated") {
  did::att_gt(
    yname      = "lemp",
    tname      = "year",
    idname     = "countyreal",
    gname      = "first.treat",
    data       = mpdta,
    est_method = method,
    control_group = control
  )
}

fit_dr <- run_attgt("dr")
```

```
fit_ipw <- run_attgt("ipw")
fit_reg <- run_attgt("reg")
```

2.1 ATT(g,t) Comparison

```
extract_att <- function(fit, label) {
  ag <- did::aggte(fit, type = "simple")
  data.frame(
    Method = label,
    ATT = round(ag$overall.att, 4),
    SE = round(ag$overall.se, 4),
    CI_lower = round(ag$overall.att - 1.96 * ag$overall.se, 4),
    CI_upper = round(ag$overall.att + 1.96 * ag$overall.se, 4)
  )
}

algo_tbl <- bind_rows(
  extract_att(fit_dr, "DR-IPW"),
  extract_att(fit_ipw, "IPW"),
  extract_att(fit_reg, "Regression")
)

kable(
  algo_tbl,
  caption = "Overall ATT by Estimation Algorithm (csdid, never-treated control)",
  booktabs = TRUE, linesep = ""
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 2: Overall ATT by Estimation Algorithm (csdid, never-treated control)

Method	ATT	SE	CI_lower	CI_upper
DR-IPW	-0.04	0.0123	-0.0641	-0.0158
IPW	-0.04	0.0125	-0.0645	-0.0154
Regression	-0.04	0.0125	-0.0644	-0.0155

2.2 Event-Study Plots by Algorithm

```
es_plot <- function(fit, title) {
  es <- did::aggte(fit, type = "dynamic")
  df <- data.frame(
    e = es$egt,
    att = es$att.egt,
    se = es$se.egt
  ) |>
    mutate(lo = att - 1.96*se, hi = att + 1.96*se)

  ggplot(df, aes(x = e, y = att)) +
    geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
    geom_vline(xintercept = -0.5, linetype = "dotted", colour = "grey50") +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2, fill = "steelblue") +
    geom_line(colour = "steelblue") +
```

```

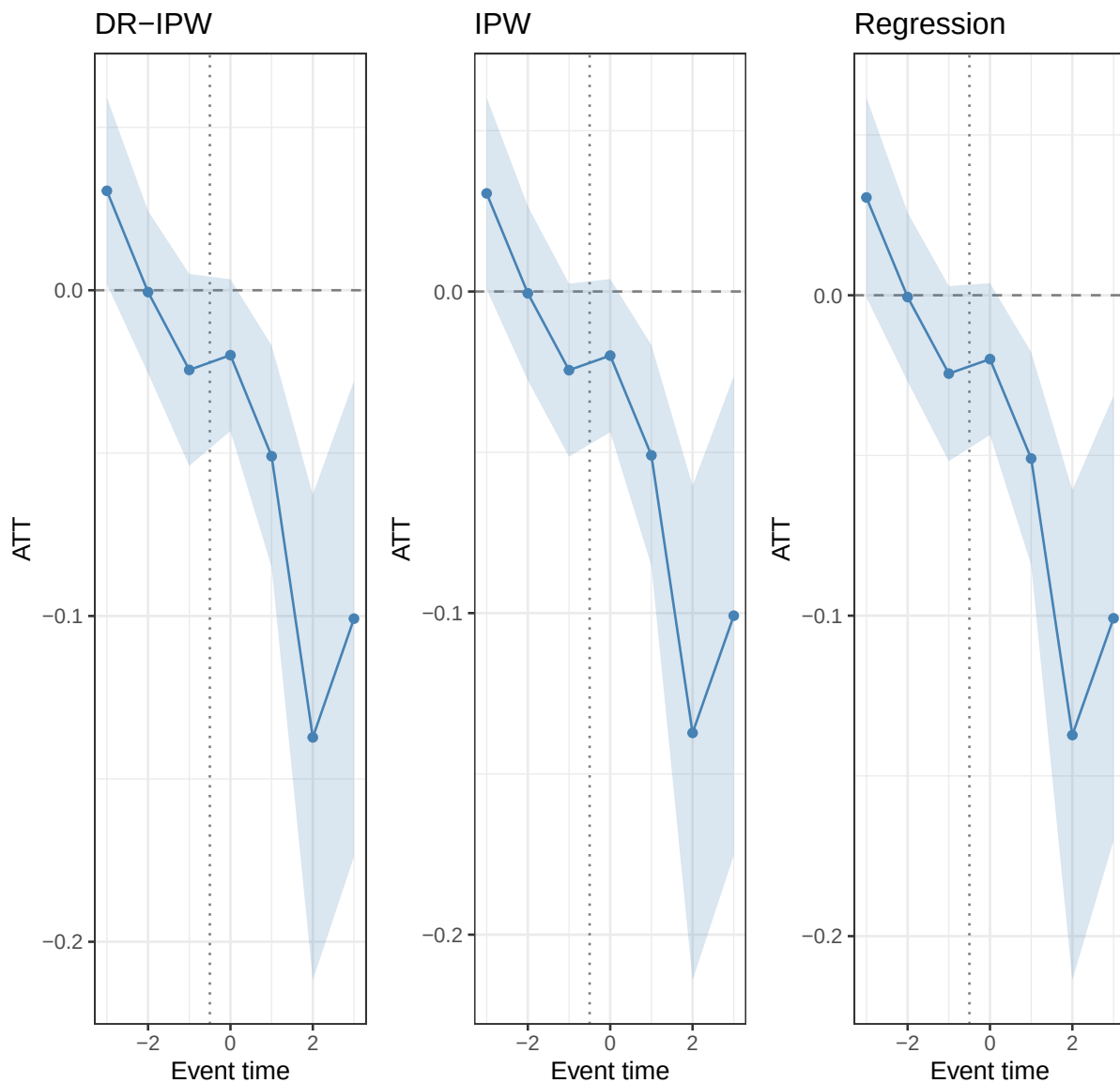
geom_point(colour = "steelblue") +
labs(title = title, x = "Event time", y = "ATT") +
theme_bw(base_size = 11)
}

p1 <- es_plot(fit_dr, "DR-IPW")
p2 <- es_plot(fit_ipw, "IPW")
p3 <- es_plot(fit_reg, "Regression")

(p1 | p2 | p3) +
  plot_annotation(title = "Event-Study: Algorithm Comparison",
                 theme = theme(plot.title = element_text(size = 13, face = "bold")))

```

Event-Study: Algorithm Comparison



Takeaway. The DR-IPW estimator is doubly robust — it remains consistent if either the propensity score or the outcome model is correctly specified. The three estimators typically agree closely on well-behaved data; divergence signals model mis-specification.

3 Part B: Comparing Base Periods (Control Group)

C&S allows two choices for the comparison group:

Option	Description
"nevertreated"	Only units never treated serve as control
"notyettreated"	Units not yet treated at time t serve as control

The choice matters when the “not-yet-treated” group may itself be partially contaminated by anticipation or when never-treated units are few.

```
fit_never <- run_attgt("dr", control = "nevertreated")
fit_notyet <- run_attgt("dr", control = "notyettreated")
```

3.1 Overall ATT Comparison

```
base_tbl <- bind_rows(
  extract_att(fit_never, "Never treated"),
  extract_att(fit_notyet, "Not yet treated")
)

kable(
  base_tbl,
  caption = "Overall ATT by Control Group Choice (DR-IPW)",
  booktabs = TRUE, linesep = ""
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 4: Overall ATT by Control Group Choice (DR-IPW)

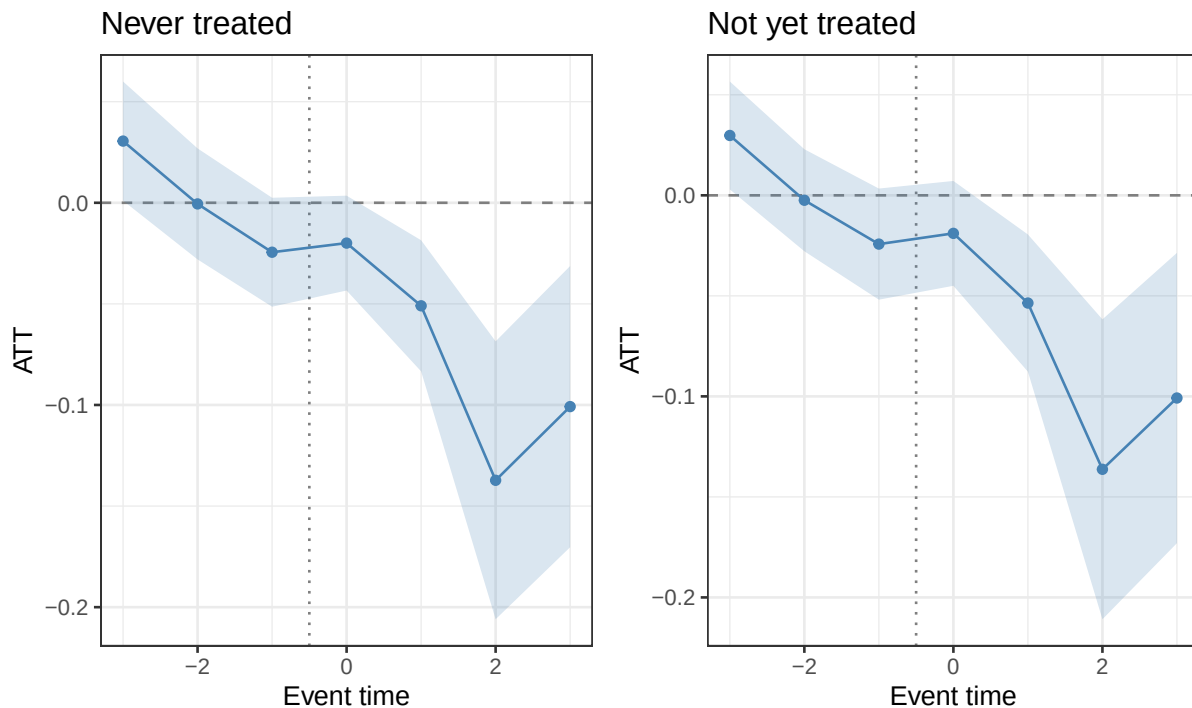
Method	ATT	SE	CI_lower	CI_upper
Never treated	-0.0400	0.0121	-0.0636	-0.0163
Not yet treated	-0.0398	0.0121	-0.0635	-0.0160

3.2 Event-Study Plots by Base Period

```
p_never <- es_plot(fit_never, "Never treated")
p_notyet <- es_plot(fit_notyet, "Not yet treated")

(p_never | p_notyet) +
  plot_annotation(title = "Event-Study: Control Group Comparison",
    theme = theme(plot.title = element_text(size = 13, face = "bold")))
```

Event-Study: Control Group Comparison

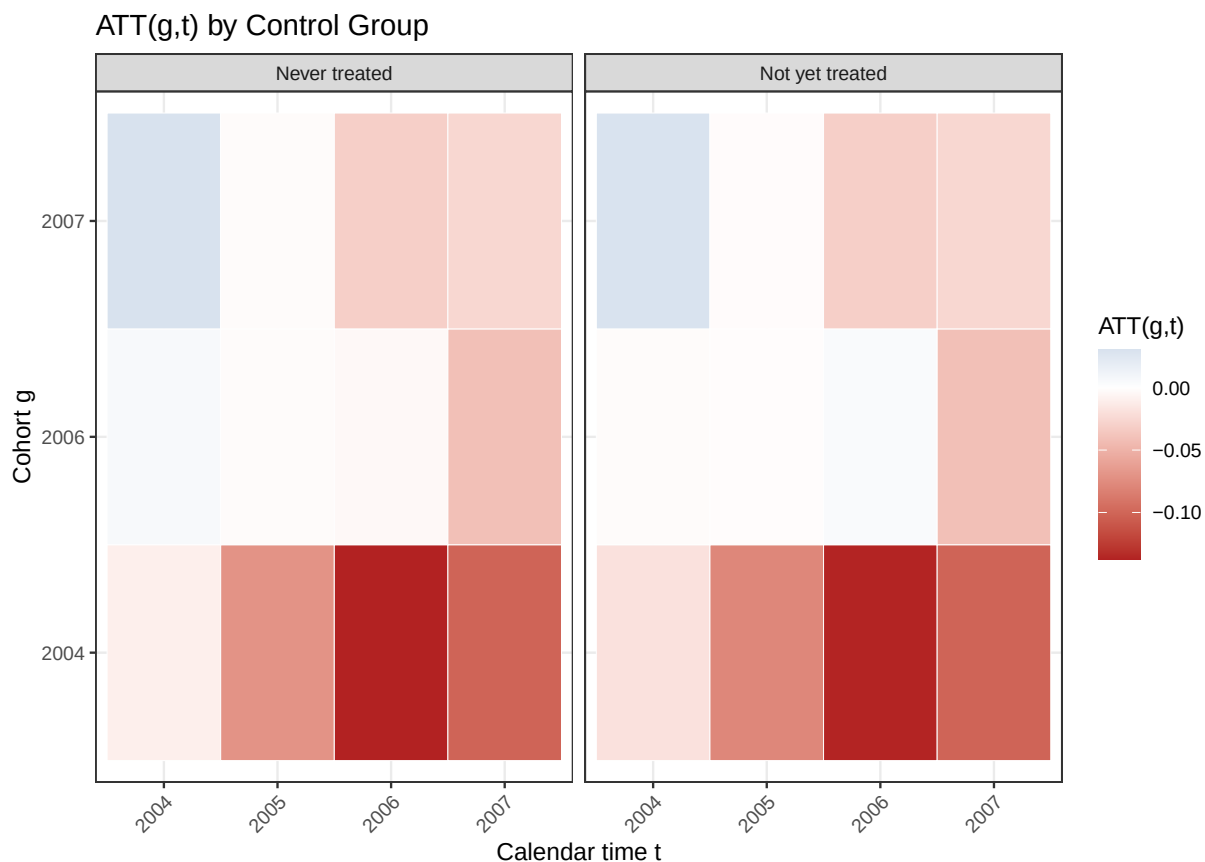


3.3 ATT(g,t) Heat Map

```
att_matrix <- function(fit, label) {
  data.frame(
    g = fit$group,
    t = fit$t,
    att = fit$att,
    source = label
  )
}

heat_df <- bind_rows(
  att_matrix(fit_never, "Never treated"),
  att_matrix(fit_notyet, "Not yet treated")
) |>
  filter(g > 0)

ggplot(heat_df, aes(x = factor(t), y = factor(g), fill = att)) +
  geom_tile(colour = "white") +
  scale_fill_gradient2(low = "firebrick", mid = "white", high = "steelblue",
    midpoint = 0, name = "ATT(g,t)") +
  facet_wrap(~source) +
  labs(title = "ATT(g,t) by Control Group",
    x = "Calendar time t", y = "Cohort g") +
  theme_bw(base_size = 10) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Takeaway. With many treated cohorts and few never-treated units, "notyettreated" gives more precise estimates but requires assuming no anticipation among later-treated units. The two choices converge when the panel is large and staggering is moderate.

4 Part C: Comparing Software Packages

4.1 R Packages

We compare three R packages against the baseline `did::att_gt()` DR result already estimated in Part A (`fit_dr`):

Package	Estimator	Notes
<code>did (att_gt)</code>	Callaway & Sant'Anna (2021)	DR-IPW, never-treated; baseline
<code>fixest</code>	Sun & Abraham (2021) via <code>sunab()</code>	TWFE-based, fast
<code>staggered</code>	Roth & Sant'Anna (2023)	Efficiency-optimised

4.1.1 `did` (baseline — already estimated)

```
att_did <- did::aggte(fit_dr, type = "simple")
cat(sprintf("did::att_gt | ATT = %.4f | SE = %.4f\n",
            att_did$overall.att, att_did$overall.se))
```

```
## did::att_gt | ATT = -0.0400 | SE = 0.0122
```

4.1.2 fixest (Sun & Abraham)

```
# Encode cohort for sunab()
mpdta2 <- mpdta |>
  mutate(cohort = if_else(first.treat == 0, Inf, as.numeric(first.treat)))

fit_sa <- feols(
  lemp ~ sunab(cohort, year) | countyreal + year,
  data = mpdta2,
  cluster = ~countyreal
)
# Aggregate: simple average over all (cohort, t) post-treatment cells
coefs_sa <- coef(fit_sa)
se_sa <- se(fit_sa)
post <- as.numeric(sub(".*::([-0-9]+)$", "\\1", names(coefs_sa)))
post_idx <- post >= 0

att_sa <- mean(coefs_sa[post_idx])
se_sa_agg <- sqrt(mean(se_sa[post_idx]^2)) # conservative; for SE use iplot()
cat(sprintf("fixest | ATT = %.4f | SE %.4f (naive avg)\n", att_sa, se_sa_agg))

## fixest | ATT = -0.0772 | SE 0.0272 (naive avg)
```

4.1.3 staggered

```
fit_stag <- staggered::staggered(
  df = mpdta,
  i = "countyreal",
  t = "year",
  g = "first.treat",
  y = "lemp",
  estimand = "simple"
)
cat(sprintf("staggered | ATT = %.4f | SE = %.4f\n",
  fit_stag$estimate, fit_stag$se))
```

```
## staggered | ATT = -0.3704 | SE = 0.1256
```

4.1.4 Summary Table

```
pkg_tbl <- tribble(
  ~Package, ~Estimator, ~ATT, ~SE,
  "did", "C&S DR-IPW", att_did$overall.att, att_did$overall.se,
  "fixest", "Sun & Abraham", att_sa, se_sa_agg,
  "staggered", "Roth & Sant'Anna", fit_stag$estimate, fit_stag$se
) |>
  mutate(
    ATT = round(ATT, 4),
    SE = round(SE, 4),
    CI_lower = round(ATT - 1.96*SE, 4),
    CI_upper = round(ATT + 1.96*SE, 4)
  )

kable(
```



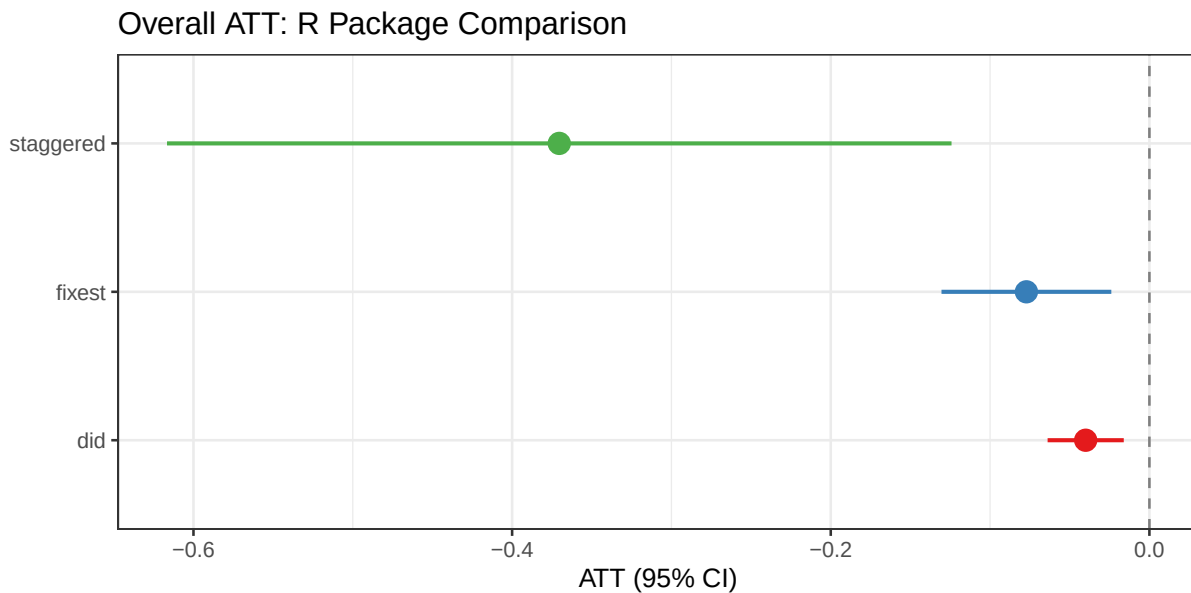
```
pkg_tbl,
caption = "Overall ATT by R Package",
booktabs = TRUE, linesep = ""
) |>
kable_styling(latex_options = c("hold_position"))
```

Table 6: Overall ATT by R Package

Package	Estimator	ATT	SE	CI_lower	CI_upper
did	C&S DR-IPW	-0.0400	0.0122	-0.0639	-0.0161
fixest	Sun & Abraham	-0.0772	0.0272	-0.1305	-0.0239
staggered	Roth & Sant'Anna	-0.3704	0.1256	-0.6166	-0.1242

4.1.5 Coefficient Plot

```
ggplot(pkg_tbl, aes(x = Package, y = ATT,
                    ymin = CI_lower, ymax = CI_upper,
                    colour = Package)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_pointrange(size = 0.8, linewidth = 0.8) +
  coord_flip() +
  scale_colour_brewer(palette = "Set1", guide = "none") +
  labs(title = "Overall ATT: R Package Comparison",
       x = NULL, y = "ATT (95% CI)") +
  theme_bw(base_size = 11)
```



4.2 Stata via Statamarkdown

The chunk below runs **Stata** code directly inside R Markdown using the **Statamarkdown** package. Requires Stata to be installed and its executable on **PATH**. Set `eval=TRUE` when running on a machine with Stata.

```

*   Install csdid if needed
*   ssc install csdid, replace
*   ssc install drdid, replace

*   Load data (export from R first, or use a Stata .dta)
*   Here we assume mpdta.dta has been written by R with:
*   haven::write_dta(mpdta, "mpdta.dta")
use "mpdta.dta", clear

xtset countyreal year

*   A. Default: DR-IPW, never-treated control
csdid lemp, ivar(countyreal) time(year) gvar(first_treat) ///
    method(dripw) notyet(0)

*   Aggregate to overall ATT
estat simple

*   B. IPW only
csdid lemp, ivar(countyreal) time(year) gvar(first_treat) ///
    method(ipw) notyet(0)
estat simple

*   C. Regression only
csdid lemp, ivar(countyreal) time(year) gvar(first_treat) ///
    method(reg) notyet(0)
estat simple

*   D. DR-IPW, not-yet-treated control
csdid lemp, ivar(countyreal) time(year) gvar(first_treat) ///
    method(dripw) notyet(1)
estat simple

*   E. Event-study plot
csdid lemp, ivar(countyreal) time(year) gvar(first_treat) method(dripw)
csdid_plot, style(rspike)
graph export "stata_eventstudy.png", replace

```

Note on Statamarkdown. Add `library(Statamarkdown)` to the setup chunk and set `stataexe = "/path/to/stata"` in `knitr::opts_chunk$set()` (or via the `STATA_EXECUTABLE` environment variable). Set `eval=TRUE` on each Stata chunk once Stata is available. Alternatively, export results from Stata as `.csv` or `.dta` and read them into R for a unified comparison table.

5 Summary Comparison

```

# Combine all R results into one master table
all_results <- bind_rows(
  # Part A: algorithms
  extract_att(fit_dr, "did · DR")      |> mutate(Dimension = "Algorithm"),
  extract_att(fit_ipw, "did · IPW")    |> mutate(Dimension = "Algorithm"),
  extract_att(fit_reg, "did · Reg")    |> mutate(Dimension = "Algorithm"),
  # Part B: base period
  extract_att(fit_never, "did · Never") |> mutate(Dimension = "Base period"),

```

```

extract_att(fit_notyet, "did · NotYet") |> mutate(Dimension = "Base period"),
# Part C: packages
data.frame(Method="did · att_gt",          ATT=att_did$overall.att,
            SE=att_did$overall.se,
            CI_lower=att_did$overall.att-1.96*att_did$overall.se,
            CI_upper=att_did$overall.att+1.96*att_did$overall.se,
            Dimension="Package"),
data.frame(Method="fixest · Sun-Abraham", ATT=att_sa, SE=se_sa_agg,
            CI_lower=att_sa-1.96*se_sa_agg, CI_upper=att_sa+1.96*se_sa_agg,
            Dimension="Package"),
data.frame(Method="staggered · R&S",      ATT=fit_stag$estimate, SE=fit_stag$se,
            CI_lower=fit_stag$estimate-1.96*fit_stag$se,
            CI_upper=fit_stag$estimate+1.96*fit_stag$se,
            Dimension="Package")
) |>
mutate(across(where(is.numeric), ~round(., 4))) |>
select(Dimension, Method, ATT, SE, CI_lower, CI_upper)

kable(
  all_results,
  caption = "Master Comparison: All Dimensions",
  booktabs = TRUE, linesep = ""
) |>
kable_styling(latex_options = c("hold_position", "scale_down")) |>
pack_rows(index = table(all_results$Dimension)) |>
column_spec(1, bold = TRUE)

```

Table 7: Master Comparison: All Dimensions

Dimension	Method	ATT	SE	CI_lower	CI_upper
Algorithm					
Algorithm	did · DR	-0.0400	0.0119	-0.0632	-0.0167
Algorithm	did · IPW	-0.0400	0.0136	-0.0666	-0.0133
Algorithm	did · Reg	-0.0400	0.0111	-0.0617	-0.0182
Base period					
Base period	did · Never	-0.0400	0.0122	-0.0639	-0.0160
Base period	did · NotYet	-0.0398	0.0124	-0.0640	-0.0156
Package					
Package	did · att_gt	-0.0400	0.0122	-0.0638	-0.0161
Package	fixest · Sun-Abraham	-0.0772	0.0272	-0.1305	-0.0240
Package	staggered · R&S	-0.3704	0.1256	-0.6167	-0.1242

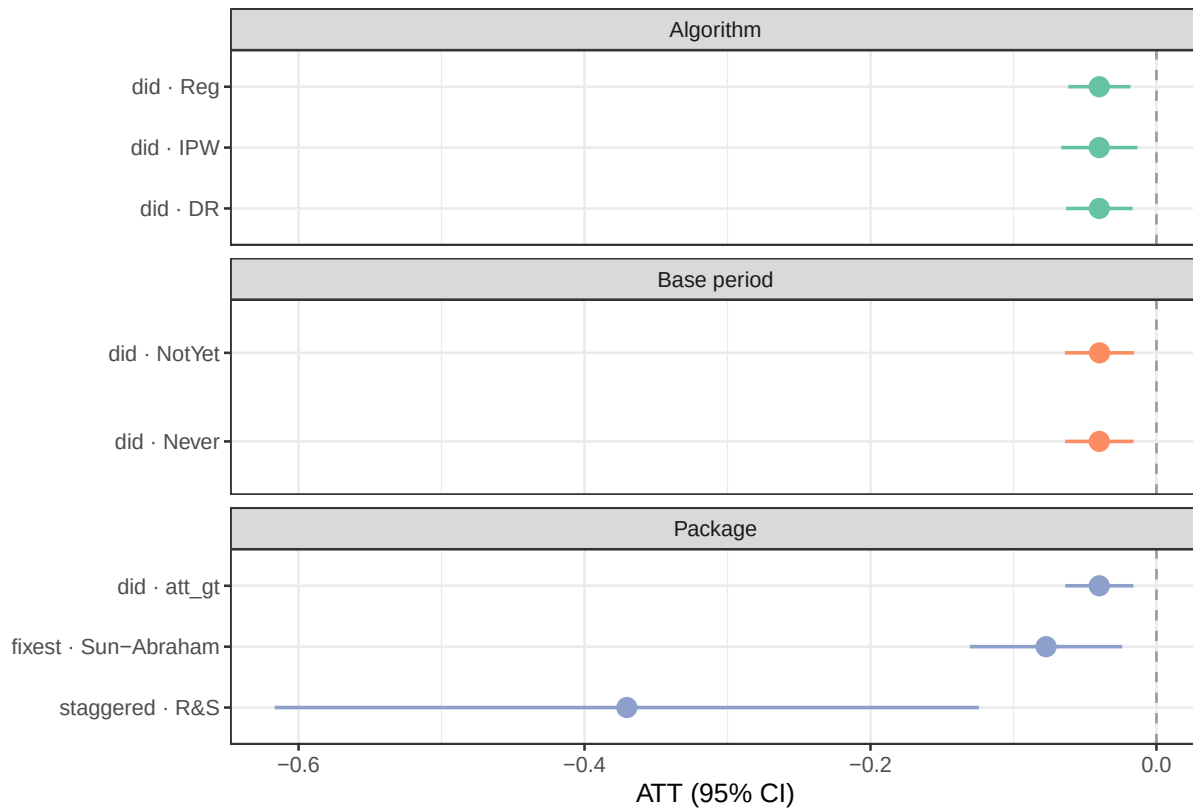
```

ggplot(all_results,
  aes(x = reorder(Method, ATT), y = ATT,
      ymin = CI_lower, ymax = CI_upper,
      colour = Dimension)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_pointrange(size = 0.7, linewidth = 0.7) +
  coord_flip() +
  facet_wrap(~Dimension, scales = "free_y", ncol = 1) +
  scale_colour_brewer(palette = "Set2", guide = "none") +
  labs(title = "Overall ATT: Master Comparison",

```

```
x = NULL, y = "ATT (95% CI)" +  
theme_bw(base_size = 11)
```

Overall ATT: Master Comparison



6 Session Info

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Asia/Taipei
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
```

```

## other attached packages:
## [1] patchwork_1.3.2 kableExtra_1.4.0 ggplot2_4.0.3 tidyr_1.3.2
## [5] dplyr_1.2.1      staggered_1.2.2 fixest_0.13.2 did_2.3.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1 viridisLite_0.4.3 timeDate_4052.112
## [4] farver_2.1.2      S7_0.2.2          fastmap_1.2.0
## [7] pROC_1.19.0.1     caret_7.0-1       digest_0.6.39
## [10] rpart_4.1.24      DRDID_1.2.3       timechange_0.3.0
## [13] lifecycle_1.0.5   dreamerr_1.5.0     survival_3.8-3
## [16] magrittr_2.0.5     compiler_4.5.1     rlang_1.2.0
## [19] tools_4.5.1       yaml_2.3.12        data.table_1.18.0
## [22] knitr_1.51         labeling_0.4.3     xml2_1.5.2
## [25] plyr_1.8.9         RColorBrewer_1.1-3 withr_3.0.2
## [28] purrr_1.2.1        numDeriv_2016.8-1.1 nnet_7.3-20
## [31] grid_4.5.1         stats4_4.5.1       future_1.70.0
## [34] BMisc_1.4.8        globals_0.19.1     scales_1.4.0
## [37] iterators_1.0.14   MASS_7.3-65        tinytex_0.59
## [40] cli_3.6.6          rmarkdown_2.31     generics_0.1.4
## [43] otel_0.2.0         rstudioapi_0.18.0  future.apply_1.20.2
## [46] bigmemory.sri_0.1.8 reshape2_1.4.4     stringr_1.6.0
## [49] splines_4.5.1      parallel_4.5.1     fastglm_0.0.4
## [52] stringmagic_1.2.0  vctrs_0.7.3        sandwich_3.1-1
## [55] hardhat_1.4.3      Matrix_1.7-3       Formula_1.2-5
## [58] listenv_0.10.1     systemfonts_1.3.1  foreach_1.5.2
## [61] trust_0.1-9        gower_1.0.2        recipes_1.3.2
## [64] glue_1.8.1         bigmemory_4.6.4     parallelly_1.47.0
## [67] codetools_0.2-20   lubridate_1.9.4     stringi_1.8.7
## [70] gtable_0.3.6       tibble_3.3.1       pillar_1.11.1
## [73] htmltools_0.5.9    ipred_0.9-15       lava_1.9.0
## [76] RcppEigen_0.3.4.0.2 R6_2.6.1           textshaping_1.0.4
## [79] evaluate_1.0.5     lattice_0.22-7      class_7.3-23
## [82] Rcpp_1.1.1-1.1     uuid_1.2-1          svglite_2.2.2
## [85] nlme_3.1-168        prodlim_2026.03.11 xfun_0.57
## [88] zoo_1.8-14         ModelMetrics_1.2.2.2 pkgconfig_2.0.3

```